# Flavio Romano

Italy | flavromano@proton.me | linkedin.com/in/flavromano | github.com/FlavRomano

Software Engineer and Technical Lead specializing in backend systems, identity & access management, and cloud-native microservices architectures.

## Experience

**Technical Lead**, NTT DATA – Pisa, IT — Sept. 2025 – Present

- Led the development of an AI-powered platform automating due diligence documentation for European sustainability regulations, coordinating 4 engineers and driving technical decisions.
- Lead a team of 5 engineers developing a sustainability platform for estimating $CO_2$ emissions from software systems, hardware infrastructure, and AI workloads. Drive technical analysis, code reviews, and platform engineering, with Spring WebFlux, PostgreSQL, and Grafana/ELK.

**Software Engineer**, NTT DATA – Pisa, IT — March 2024 – Sept. 2025

- Contributed to the design and implementation of OAuth 2.0, OpenID Connect, and SAML authentication solutions for a leading Italian telecommunications provider, enhancing a customized Keycloak deployment and enabling adoption of the Phantom Token architecture.
- Developed Spring Boot microservices supporting authorization workflows, including token introspection, redirect management, and JWT/JWE processing.

**Software Engineer Intern**, NTT DATA – Pisa, IT — Dec. 2023 – March 2024

- Developed a Java microservice for the GreenIT ecosystem to collect hardware metrics and estimate Software Carbon Intensity (SCI) and $CO_2$ emissions on Google Cloud Platform infrastructures.
- Designed a configurable monitoring connector that later became the foundation of my Bachelor's thesis.

## Technical Skills

**Languages:** Java, Python, Go, TypeScript, SQL, Bash, C
**Backend & Cloud:** Spring Boot, Spring WebFlux (Reactive), FastAPI, PostgreSQL, Kubernetes, AWS, GCP, RabbitMQ, Redis, Terraform, Grafana, ELK Stack
**Security:** OAuth 2.0, OpenID Connect, SAML, Keycloak, RBAC

## Languages

**Italian:** Native
**English:** C1 – Advanced

## Projects

**Encoding Classification using LSTM** (github) — Apr. 2026 – May 2026

- Developed a PyTorch-based sequence classification model to automatically identify the encoding algorithm (Caesar Cipher, ROT13, MD5) applied to a text string.
- Built and trained a 230k-parameter LSTM on 72,000+ samples, achieving 97.99% test accuracy, with automated training/inference workflows for real-time prediction.

**Coinquilining** (github) — Aug. 2023 – Dec. 2023

- Built from scratch as the final project for the "Web App Development" course (top grade, 30/30): an installable, offline-capable PWA for shared households, covering fridge/pantry inventory, shopping lists, cleaning rotations, and expense splitting/balances.
- Implemented expiration-tracking notifications, ingredient-based recipe search via Puppeteer scraping, and a fair cleaning-rotation algorithm rendered as a drag-and-drop calendar. Tech stack: SvelteKit, TypeScript, Supabase (Postgres/Auth), Tailwind CSS, Puppeteer.

## Education

**University of Pisa**, Master of Science (MSc) in Artificial Intelligence — Sept. 2025 – Present
**University of Pisa**, Bachelor of Computer Science — Sept. 2020 – May 2024